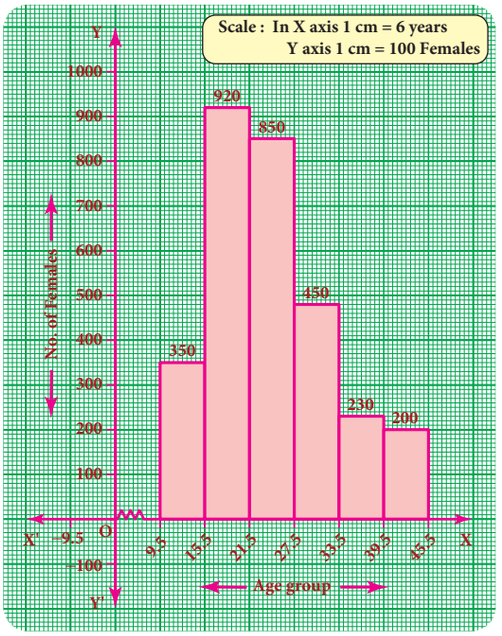
Solution:

The given distribution is discontinuous. If we represent the given data as it is by a graph we shall get a bar graph, as there will be gaps in between the classes. So, convert this into a continuous distribution using the adjustment factor 0.5.

The first class interval can be written as 9.5-15.5 and the remaining class intervals are changed in the same way. There are no changes in frequencies.

The new continuous frequency table is



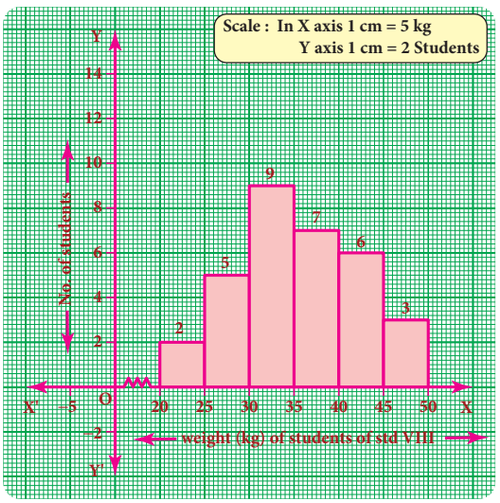
Age group	9.5-15.5	15.5-21.5	21.5-27.5	27.5-33.5	33.5-39.5	39.5-45.5
No of females	350	920	830	480	230	200

Example 6.8

Observe the given histogram and answer the following questions

Hint: Under weight: less than 30 *kg;* Normal weight: 30 to 45 *kg;* Obese: More than 45 *kg*

1. What information does the histogram represent?
2. Which group has maximum number of students?
3. How many of them are under weight?
4. How many students are obese?
5. How many students are in the weight group of 30-40 kg?



Solution:

1. The histogram represents the collection of weight from std VIII.
2. There are maximum 9 students in 30-35 kg weight.
3. There are $7(=2+5)$ students who are under weight.
4. There are 3 students who are obese.
5. There are $16(=9+7)$ students in the 30-40 kg weight group.

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